Twenty-Metre WSPR Propagation Observed from a Mobile 200 mW Beacon in Southern Finland

Diurnal, geographic, seasonal and space-weather patterns, 2026-05-08 to 2026-11-06

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WSPR Logger project note — version 1.0 — 6 November 2026 — `github.com/oh2gax/wspr_logger`

Abstract

A 200 mW WSPR beacon on 14 MHz (callsign OH2GAX, Diamond monoband whip on a car roof, home locator KP20jg) has been logged continuously by the WSPR Logger software. This note analyses 166 calendar days (2026-05-08 to 2026-11-06): 18 867 poll snapshots, 1 095 975 individual reception reports from 1 747 distinct receiving stations, together with the solar and ionospheric indices recorded at every poll (SFI, K and A index, X-ray class, Juliusruh MUF and foF2). The band is effectively closed between about 00 and 04 UTC (median 6 stations or fewer) and carries a median of about 78 stations between 08 and 16 UTC. 94.5 % of the receiving stations are in Europe and 87 % lie between south and west-south-west of the transmitter. At least one station farther than 6 000 km was heard on 131 of 166 days (79 %), and one farther than 10 000 km on 53 days; the record is 15 619 km (VK7JJ, 2026-06-23), i.e. 78 095 km per watt. Long-distance windows are concentrated to the early morning (North America around 11 UTC, 10–13 UTC) and late afternoon (Africa and Oceania, 16–19 UTC). After removing the diurnal rhythm, reporting is strongly tied to the Juliusruh MUF (index 27 for MUF < 12 MHz versus 144 at 20–24 MHz) and weakly and negatively to the geomagnetic K-index (correlation -0.16); the 10.7 cm solar flux alone explains nothing ($r = 0.06$). Across the six complete months the median daytime reporting rose from 32 stations in July to 110 in October while SFI fell, so season and MUF, not SFI, dominate. The worst day of the record (4 July, K = 7, A = 74, 10 stations against a typical 80) followed an M-class flare period with SFI above 195.

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1 Introduction

The Weak Signal Propagation Reporter (WSPR) protocol turns every amateur transmitter into a propagation probe: a two-minute, 6 Hz-wide signal is decoded by volunteer receivers around the world and each decode is stored with the receiver position, signal-to-noise ratio (SNR) and distance [1, 3]. A 200 mW beacon is a modest probe, but because the same transmitter transmits on the same schedule every day, the stream of receptions is a long, homogeneous time series of the 20 m band as seen from one place.

The WSPR Logger started as a live dashboard for such a beacon: it polls the public WSPR database `wspr.live` [2] every ten minutes, draws the reporters on a map, keeps a log of the farthest reporter per transmission and records the solar conditions at the time. A second database, the *replay database*, stores every poll snapshot in full. The present note is the first systematic analysis of that archive. It asks four questions:

(a) When is the band open, and when are long-distance paths available?

(b) In which directions and at which distances does a mobile 200 mW beacon in Finland actually get heard?

(c) How does reporting change over the season?

(d) How much of the variation is explained by the space-weather indices the logger records?

The note is deliberately descriptive. Where a number can be given with a confidence interval it is; where the design of the data collection limits what can be concluded, this is stated (Section 5).

2 Station, data and definitions

2.1 Transmitting station

The transmitter is a ZachTek WSPR desktop transmitter [4] with built-in GPS (used for timing and for the position), producing about 200 mW (23 dBm) into 50 Ω. The antenna is a Diamond monoband 14 MHz whip on a magnetic mount on the roof of a VW Touran. The home locator is KP20jg (60.27°N, 24.79°E), but the car moves; 84 % of the 15 942 logged transmissions carry the home locator and 72 different six-character locators occur in total (the mobile aspect is the subject of a companion paper on position reporting). One WSPR transmission is made every ten minutes.

2.2 Reception data

Every ten minutes (at :08, :18, . . .) the logger queries `wspr.live` for the receptions of the last 45 minutes and stores, for the current *snapshot*, every station that has reported the beacon within the preceding 60 minutes: callsign, Maidenhead grid, SNR (dB, 2 500 Hz reference bandwidth) and great-circle distance to the transmitter. This gives 18 867 snapshots on 166 calendar days; because the logger was occasionally offline the snapshots cover about 131 days of 24 hours (79 % of the period). The replay database holds 1 095 975 rows. A second, older database stores one record per heard transmission (15 942 records on 168 days) with the number of reporters and the largest distance.

2.3 Solar-terrestrial data

At every poll the logger stores the 10.7 cm solar flux (SFI), the planetary K and A indices, the GOES X-ray class, and the maximum usable frequency (MUF, 3 000 km reference) and critical frequency foF2 of the Juliusruh ionosonde (54.6°N, 13.4°E) [5, 6]. Indices come from NOAA SWPC-derived feeds [7, 8]. Readings of zero in SFI or MUF are feed failures (76 snapshots) and are treated as missing. foF2 is available for only 16 % of the snapshots (2 947) because the earlier foF2 source was unreliable; it is not used quantitatively here.

2.4 Definitions

Reporters n_t
number of distinct stations that reported the beacon in the 60 minutes before snapshot t.

Expected reporters $\bar{n}_h$
mean of n_t over all snapshots in UTC hour h (the typical value for that hour).

Index
$100 \sum n_t / \sum \bar{n}_{h(t)}$ over a set of snapshots. A value of 100 means "typical for that hour of the day"; this removes the strong day–night rhythm, which would otherwise masquerade as a solar effect.

DX window
snapshot in which at least one reporter was farther than 6 000 km. DX share is the percentage of snapshots that are DX windows.

Daytime
08–16 UTC (10–18 or 11–19 local time, depending on daylight saving).

2.5 Quality control

Each (callsign, grid) pair was compared with the distance the logger stored: a pair is rejected if the distance computed from the receiver's grid differs from the reported distance by more than 500 km. Of 2004 pairs, 1 was rejected: `G8LZI/2` (grid JO02gl) is listed at 11 643 km although its grid lies 1 687 km from the transmitter; its 1 033 rows were dropped from every statistic below. Without this step the farthest "reporter" of the whole record would have been spurious. All station and distance results use the cleaned data.

3 Methods

Confidence intervals. Proportions of snapshots (DX shares) carry 95 % Wilson intervals. Indices and correlations carry 95 % day-block bootstrap intervals (2 000 resamples of whole days with replacement), because consecutive snapshots share 50 of their 60 minutes and are strongly autocorrelated; treating them as independent would make every interval far too narrow.

Regions. Receiving stations were assigned to a region from the centre of their four-character grid. **Sectors.** Bearing from the transmitter home locator is divided into eight 45° sectors. **Distance bands.** 0–1 500, 1 500–3 000, 3 000–6 000, 6 000–10 000 and >10 000 km. A *station* statistic uses the median over the snapshots in which that station appears.

Space-weather bins. MUF, K and A are binned and, within each bin, the index and DX share are computed over all snapshots in the bin. Bins with fewer than 30 snapshots are omitted. Correlations use the per-snapshot index $n_t/\bar{n}_{h(t)}$ (relative reporting) and the raw n_t for comparison.

Reproducibility. All figures and numbers in this note are generated by `analyze.py` from the two databases; the LaTeX macros in the text are written by the script, so no value is typed by hand (Appendix A).

4 Results

4.1 Overview

The record contains 18 867 snapshots from 2026-05-08 to 2026-11-06, with a mean of 58.0 and a median of 50 reporters per snapshot; 1 747 distinct callsigns appear (2 003 callsign–grid pairs). 79 % of days (131 of 166) had at least one reporter beyond 6 000 km, 53 days (32 %) beyond 10 000 km. 14.7 % of all snapshots are DX windows. Figure 1 shows the structure at a glance: a regular daytime band of strong reporting, a dead band in the small hours, week-long dropouts of the logger, and a clear brightening in September–October.

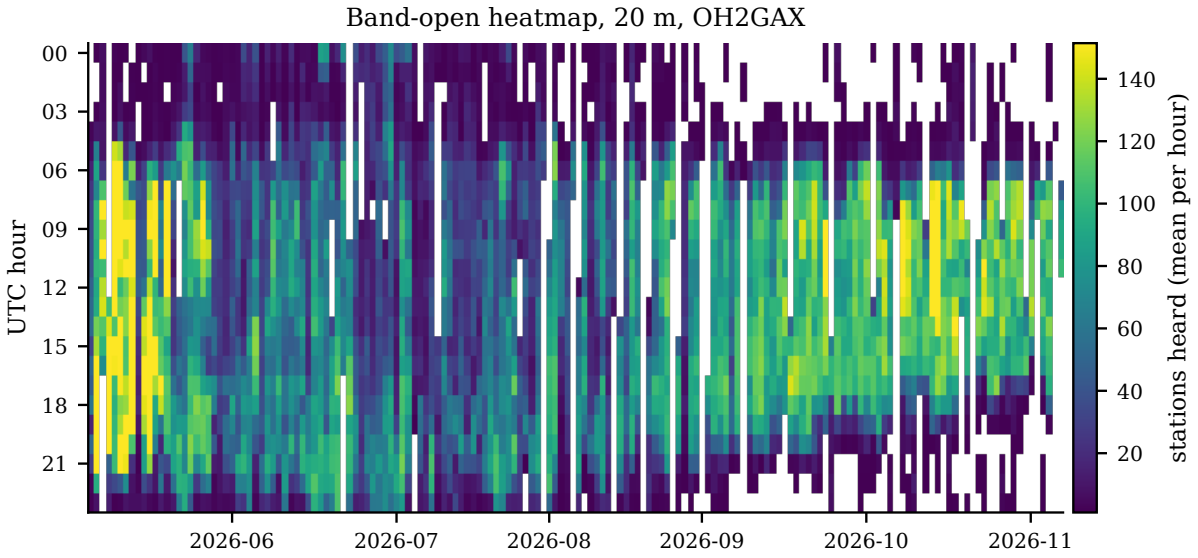


Figure 1. Band-open heatmap for 20 m: mean number of stations heard per UTC hour for each day (colour clipped at the 97th percentile). White columns are periods when the logger was not running.

4.2 Diurnal pattern

Figure 2a shows the typical day. Between 00 and 04 UTC the median is ≤6 stations; reporting rises steeply after 05 and the median is at least 50 stations from 7 to 20 UTC; from 08 to 16 UTC it is about 78 (maximum hourly median 80). The interquartile range is wide, on average 41 to 106 stations between 08 and 16 UTC, so day-to-day variability is large compared with the typical value.

The probability of a long-distance window is *not* highest when the band is busiest (Figure 2b). The share of snapshots with a reporter beyond 6 000 km peaks at 24 % around 11 UTC but is only 3 % at 8 UTC; 21 % at 05 UTC and 13 % at 19 UTC. A reporter beyond 3 000 km is available in up to 56 % of snapshots at 20 UTC and in only 13 % at 14 UTC. A plausible reading is that around noon the F2 layer carries the short and medium paths into Europe while the long paths, which need the layer to be illuminated at both ends or along a grey line, are least likely; the morning and evening maxima coincide with the periods when the path to North America (morning) and to Africa and Oceania (afternoon and evening) passes through twilight. The logger data cannot test this mechanism; they only document the timing.

4.3 Geography: where the beacon is heard

Table 1 and Figure 3 show that the footprint is overwhelmingly European: 1893 of 2 003 stations (94.5 %). North America contributes 53 stations (2.6 %), Africa 28, Oceania 15 and South America 1; Asia only 2. Measured by bearing, 1345 stations lie in the south-west sector (225–270°), 401 in the south sector (180–225°), 86 to the west and 77 to the south-east; the east and north together hold only about 2 % and 2 % of the stations. The receiver network is densest in Central and Western Europe, which are also the

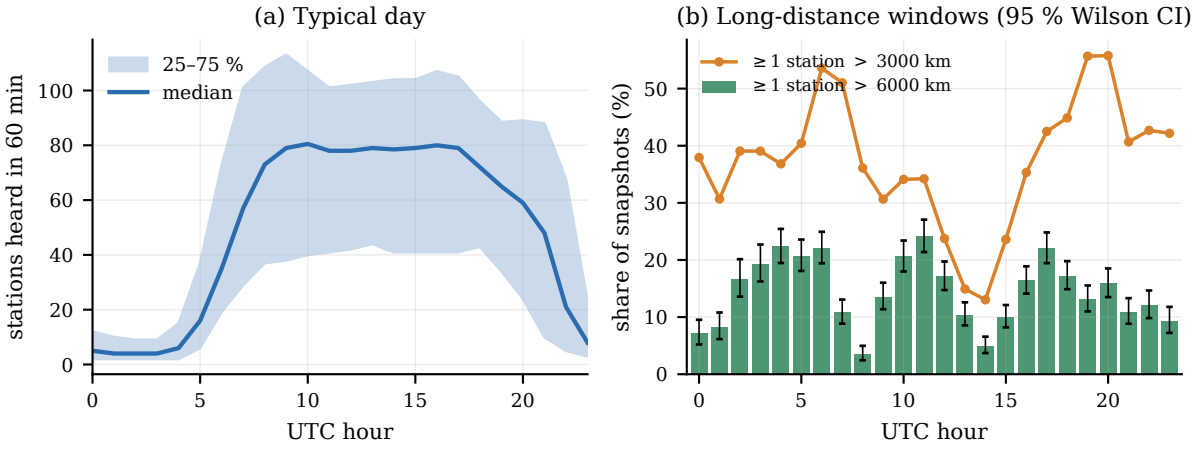


Figure 2. (a) Typical day: median and interquartile range of the stations heard in 60 minutes by UTC hour. (b) Share of snapshots with at least one station beyond 6 000 km (bars, 95 % Wilson intervals) and beyond 3 000 km (line).

directions of the best 20 m paths from Finland, so this is a joint property of propagation and receiver density, not of propagation alone.

The reporting is also concentrated on a few stations: 73 stations account for half of all reports (the three most frequent are G4ZFQ, ON5KQ, DK6UG; the top station appears in 14 547 snapshots). These are mostly continuously running receivers, so the network response to a change in propagation is dominated by a small set of fixed receivers.

Table 1. Receiving stations by region (unique callsign–grid pairs, quality-controlled).

Region of station	Stations	Share (%)	Median dist. (km)	Max. dist. (km)	Best SNR (dB)
Europe	1 893	94.5	1 632	3 538	+24
North America	53	2.6	6 793	9 102	-7
Africa	28	1.4	3 185	9 877	-5
Oceania	15	0.7	15 087	15 619	-10
South America	1	0.0	13 181	13 181	-5
Asia	2	0.1	2 331	2 470	-9
Other / unknown	11	0.5	3 691	3 785	-14

4.4 Distance and signal strength

Over all reports the median SNR is -19 dB with quartiles at -24 and -14 dB (5th percentile -28 dB); the lowest value in the archive is -35 dB and the highest +24 dB, measured at `DK6UG` at 1 592 km (Table 2). SNR falls with the logarithm of distance, but weakly (correlation -0.24 between $\log_{10} d$ and the station median SNR): the median is −19 dB below 1 500 km, −22 dB for 1 500–6 000 km and only −24 to −25 dB beyond 6 000 km. The long-distance stations are also the most uniform (interquartile range of 3 dB or less) and lie close to the lower decoding limit of the archive, about −33 dB, so the long paths operate with little margin and a few dB of extra attenuation removes them. The weak distance dependence is what one expects for a single F2 hop whose loss is dominated by absorption and focusing rather than by free-space spreading; the logger data do not test this explanation.

4.5 Long-distance windows by region

Long-distance reports (>6 000 km) come from three regions. North America reported on 96 days, with a morning maximum around 11 UTC (up to 31 % of days at that hour) and a secondary peak in the early

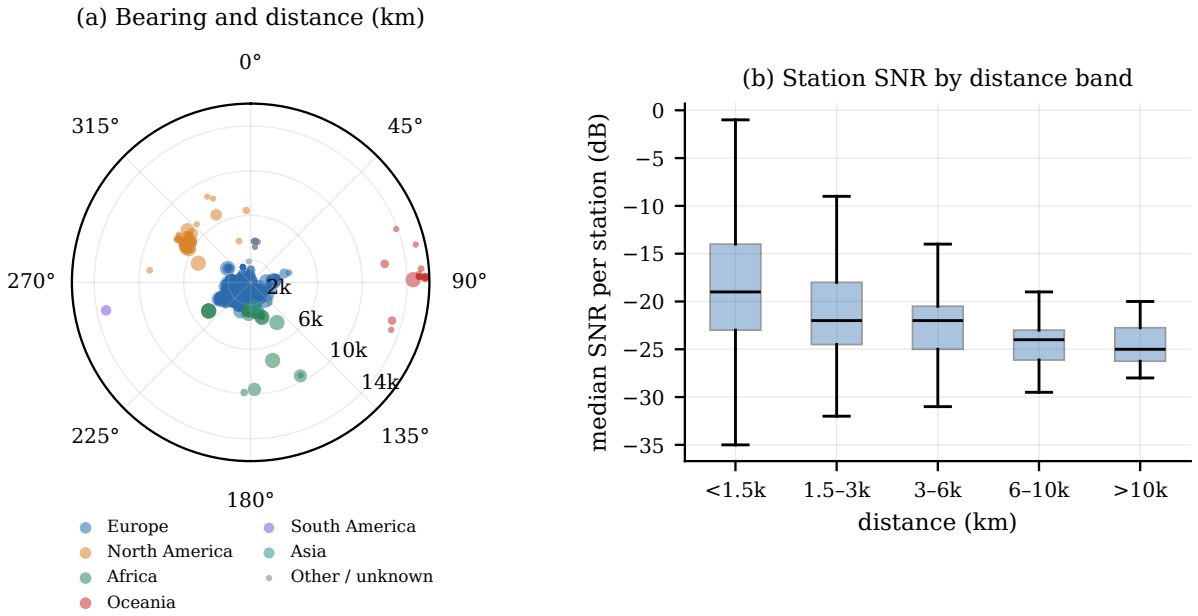


Figure 3. (a) Position of every receiving station by bearing from the transmitter and distance (dot size grows with the number of snapshots in which the station was heard; colour = region). (b) Median SNR per station by distance band (boxes: quartiles; whiskers: 1.5 interquartile ranges).

Table 2. Station SNR by distance band (median over the snapshots in which a station was heard).

Distance (km)	Stations	Median SNR (dB)	Q1 (dB)	Q3 (dB)	Best SNR (dB)
< 1 500	603	-19	-23	-14	+15
1 500–3 000	1 277	-22	-24	-18	+24
3 000–6 000	51	-22	-25	-20	-5
6 000–10 000	56	-24	-26	-23	-7
> 10 000	16	-25	-26	-23	-5

hours. Africa reported on 57 days, concentrated in the afternoon and evening (16–19 UTC; maximum 21 % of days at 17 UTC). Oceania reported on 53 days, mainly in the late afternoon and with a smaller early-morning window (14 % maximum at 17 UTC). South America contributed on 1 day only. The farthest of each region is in Table 3.

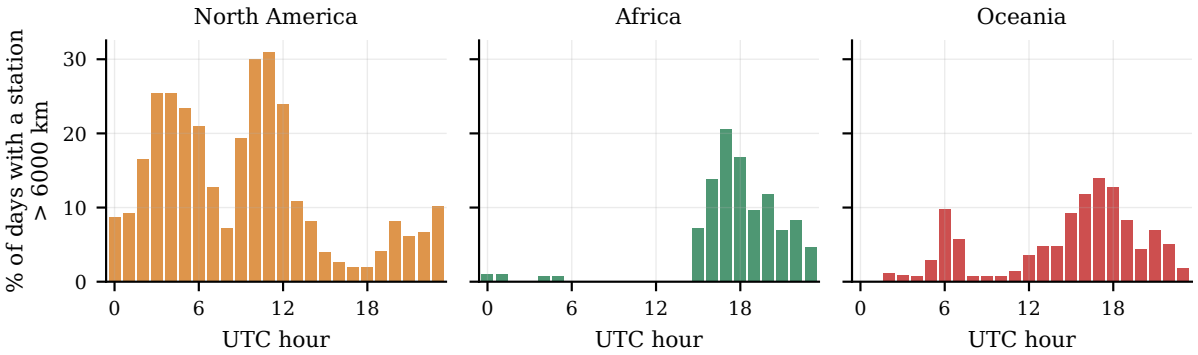


Figure 4. Percentage of days with at least one station of the region beyond 6 000 km in each UTC hour, for the three regions that produced long-distance reports.

Table 3. Days with at least one station beyond 6 000 km, by region.

Region	Days	Best 3-h window (UTC)	Peak share (%)	Farthest (km)	Station, date
North America	96	10–13	31	9 102	YV5RAB, 2026-09-24
Africa	57	16–19	21	9 877	ZS3Y, 2026-06-25
Oceania	53	16–19	14	15 619	VK7JJ, 2026-06-23
South America	1	–	–	13 181	CE3AM, 2026-06-28

Share = percentage of days with data in that UTC hour on which at least one station of the region beyond 6 000 km was heard.

4.6 Seasonal variation

Monthly daytime reporting (Table 4, Figure 5) shows a pronounced seasonal pattern that is *not* the one the solar flux alone would predict. The median daytime count fell from 87 in May to 32 in July, then rose to 110 in October (102 in November so far, from six days). The mean SFI went the other way: 121 in May, 139 in June and 133 in July, down to 110 in October. The correlation of the monthly median reporting with monthly mean SFI is -0.87 and with the monthly median Juliusruh MUF 0.91 (six complete months only, so these are descriptive, not statistical, values).

Two effects probably contribute and cannot be separated in this record. First, the mid-latitude F2 layer has a well-known seasonal anomaly: for the same solar activity the daytime critical frequency is higher in the winter half-year than in summer, which raises the MUF in autumn (here from a daytime median of 19.8 MHz in May and 18.1 MHz in July to 24.4 MHz in October and 25.9 MHz in November). Second, the receiving network itself changes: the number of distinct stations that heard the beacon fell from 665 in May to 515 in July and rose to 766 in October, which is consistent with fewer attended and automatic receivers during the northern-hemisphere summer. Without an independent reference beacon, neither explanation can be ruled out.

4.7 Space weather

Figure 6 and Tables 5–7 relate the hour-adjusted reporting index and the DX share to the three indices recorded at each poll.

MUF. The Juliusruh MUF is the strongest single predictor. At MUF below 12 MHz the index is only 27 and the DX share 4 %; at 20–24 MHz the index is 144 and the DX share 19 %. The raw correlation between reporters and MUF is 0.70 (95 % CI 0.67–0.74), but much of that is the common diurnal rhythm; after

Table 4. Monthly summary. Stations = distinct callsigns heard in the month.

Month	Days	Snapshots	Stations	Median stations 08–16 UTC	Median MUF (MHz)	Mean SFI
2026-05	24	3 556	665	87	19.8	121
2026-06	28	3 762	600	49	18.8	139
2026-07	26	3 370	515	32	18.1	133
2026-08	26	2 737	642	65	19.9	117
2026-09	28	2 597	707	99	22.5	106
2026-10	28	2 532	766	110	24.4	110
2026-11	6	313	331	102	25.9	99

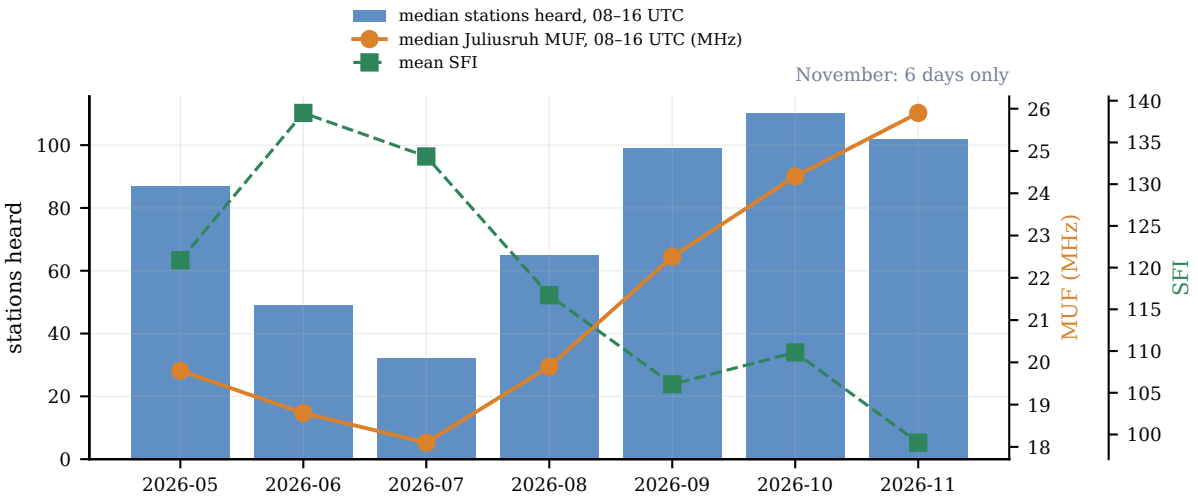


Figure 5. Monthly median number of stations heard between 08 and 16 UTC (bars), median Juliusruh MUF over the same hours (circles) and mean SFI (squares). November comprises six days.

removing it the correlation with relative reporting is 0.43 (0.37–0.48). The relationship is therefore real but moderate: a high MUF is necessary rather than sufficient.

Geomagnetic activity. Reporting falls with the K-index, but gently: the index is 113 at $K = 0$ and 88 at $K \geq 5$; the correlation is -0.16 (-0.22 to -0.10). The DX share is far more sensitive: 22 % at $K = 0$, 19 % at $K = 1$ and only 7 % on average for $K \geq 3$. The A-index gives the same picture over the day: for $A < 8$ the DX share is 20 % (index 111), for $A \geq 30$ it is 2.7 % and the index 66. Disturbed conditions thus mostly remove the weakest, longest paths first, which is the expected behaviour of an auroral-zone path at 60°N.

Solar flux. The solar flux has no detectable effect on hourly reporting on its own ($r = 0.06$, CI -0.05 to 0.15; the raw value is -0.13). This is not a statement that the Sun is irrelevant: SFI changed only between about 89 and 203 during the record, strong flux comes with flares and storms (Section 4.8), and SFI is a daily number while reporting responds to the hour-to-hour ionosphere.

Table 5. Reporting versus Juliusruh MUF. Index: stations heard relative to the typical value for the same UTC hour (100 = typical); DX share: snapshots with a station beyond 6 000 km. Confidence intervals by day-block bootstrap.

MUF (MHz)	Snapshots	Mean stations	Index	95 % CI	DX share (%)	95 % CI
< 12	1 549	6	27	21–33	4	2–7
12–16	3 494	13	38	34–42	10	7–13
16–20	7 534	52	81	75–88	15	13–18
20–24	4 710	103	144	135–153	19	16–23
24–28	1 160	110	142	130–155	20	14–27
≥ 28	335	100	126	113–138	17	8–29

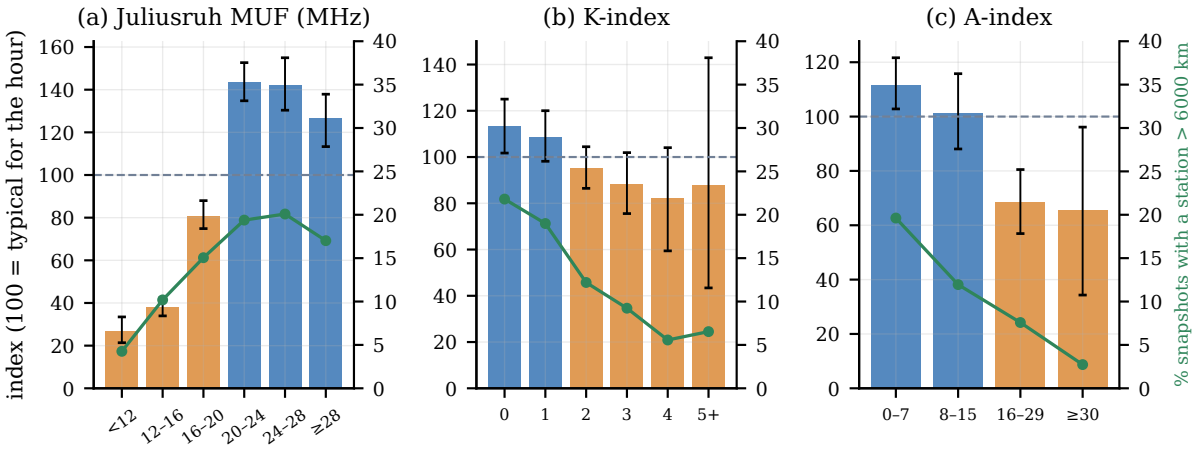


Figure 6. Hour-adjusted reporting index (bars, left axis, 100 = typical for the UTC hour; 95 % day-block bootstrap intervals) and DX share (green line, right axis) by (a) Juliusruh MUF, (b) K-index and (c) A-index. Blue bars are at or above the typical level, orange below.

Table 6. Reporting versus the K-index (day-block bootstrap intervals).

K-index	Snapshots	Mean stations	Index	95 % CI	DX share (%)	95 % CI
0	2 325	64	113	102–125	22	16–28
1	6 240	65	109	98–120	19	16–22
2	5 830	55	95	86–104	12	10–14
3	3 174	50	88	76–102	9	6–13
4	880	45	82	59–104	6	0–12
5+	398	51	88	43–143	7	1–15

Table 7. Reporting versus the A-index (day-block bootstrap intervals).

A-index	Snapshots	Mean stations	Index	95 % CI	DX share (%)	95 % CI
0–7	9 178	65	111	103–122	20	17–22
8–15	6 200	59	101	88–116	12	9–15
16–29	2 757	39	69	57–81	8	5–10
≥ 30	732	41	66	34–96	3	0–7

4.8 Case study: 24 June – 9 July

The most disturbed fortnight of the record is instructive (Table 8, Figure 7). Solar flux rose from 131 on 24 June to above 195 on 30 June–3 July with M-class flares on several days. Reporting followed no simple pattern: 2 and 3 July had the best days of the fortnight (about 93 and 70 stations, with K = 2–3) while 25 June (K = 5, A = 26) and 1 July (K = 4, A = 18) were weak. The worst day of the entire record was 4 July: K = 7, A = 74, and only 10 stations in the daytime window against the typical 80. The disturbance persisted: 5 July (K = 5, A = 74) had 13 stations, and 6 July, with K down to 3 and A = 18, still only 10; reporting recovered to 28 on 7 July and to 67 on 9 July. The pattern, a strong single-day collapse followed by a recovery lag of two to three days, is the classic ionospheric storm signature at high geomagnetic latitudes: it is the K-index and the A-index, not the X-ray class or the flux, that mark the days that failed.

Table 8. The period 24 June – 9 July 2026: space weather and reporting (daytime mean).

Date	SFI	Max K	Max A	Max X-ray	Stations 08–16 UTC
2026-06-24	131	3	15	C	37
2026-06-25	139	5	26	C	16
2026-06-26	157	4	26	C	24
2026-06-27	166	3	12	C	15
2026-06-28	188	2	8	C	39
2026-06-29	187	1	8	C	27
2026-06-30	196	4	18	M	40
2026-07-01	203	4	18	M	16
2026-07-02	201	2	17	M	93
2026-07-03	201	3	12	M	70
2026-07-04	184	7	74	M	10
2026-07-05	159	5	74	M	13
2026-07-06	138	3	18	C	10
2026-07-07	123	2	7	C	28
2026-07-09	115	4	16	C	67

4.9 Records

The eight farthest stations (Table 9) include Oceania and North America; the record is VK7JJ at 15 619 km on 2026-06-23, which corresponds to 78 095 km per watt of transmitter output. Three of the eight rows (`VK7JJ`, `VK7FLYN`, `VK7FLYN/2`) share one grid square and are probably one receiving site using several callsigns. Of the 16 stations beyond 12 000 km, 15 are in Australia and one in South America (a single day in June).

5 Discussion

What the data support. The diurnal structure, the European concentration and the existence and timing of the North American, African and Oceanian windows are all robust: they rest on hundreds of days or thousands of snapshots and are not sensitive to the quality-control step. The dependence of reporting

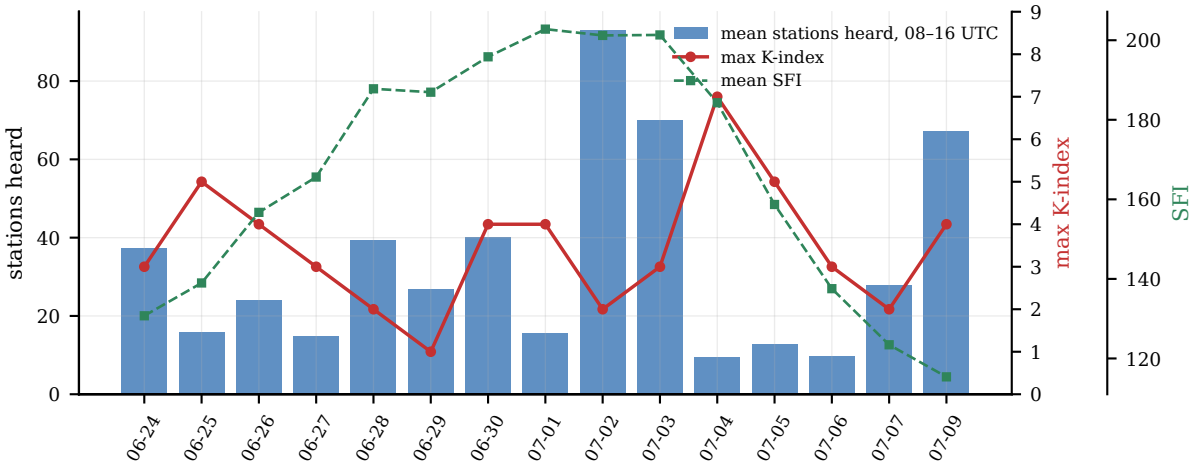


Figure 7. Daytime reporting (bars) with maximum K-index (red) and mean SFI (green) for 24 June – 9 July 2026 (8 July had no complete daytime data).

Table 9. The eight farthest stations (one row per station, quality-controlled).

#	Station	Region	Grid	Distance (km)	When (UTC)
1	VK7JJ	Australia	QE38lr	15 619	2026-06-23 21:48
2	VK7FLYN	Australia	QE38lr	15 619	2026-09-24 06:08
3	VK7FLYN/2	Australia	QE38lr	15 619	2026-09-30 05:48
4	VK7EKA	Australia	QE38kv	15 596	2026-10-13 06:39
5	VK3AG	Australia	QF31cs	15 319	2026-05-26 16:38
6	SWL/VK3	Australia	QF12wb	15 165	2026-06-14 21:38
7	VK2EFM	Australia	QF56oq	15 156	2026-09-21 05:39
8	VK3CMZ	Australia	QF13xa	15 087	2026-07-21 16:08

on MUF and on K/A is also robust in sign and, with day-block intervals, in size. The seasonal pattern is real in the data but cannot be attributed.

Limits.

- *One path, one band, one season.* The archive covers the 2026 summer season, from 2026-05-08; the transmitter is on 14 MHz only. Earlier seasons of the same station were not archived in this form and the car-mounted antenna is not used in winter (snow, low temperature), so no winter data exist. Seasonal statements are therefore about May–November and cannot be extrapolated to the full year.
- *Receiver network.* Reporting counts depend on how many receivers are listening. The 73 most frequent stations account for half of all reports, so a station going off-air shifts the counts without any change in propagation. The sparsely covered directions (north, east) say nothing about propagation there.
- *Mobile transmitter.* The car moves; 84 % of the logged transmissions come from the home square. Position-dependent effects (antenna surroundings, local noise at the receiving end of a path) are not separated. The companion paper on position reporting examines the movement itself.
- *Overlapping windows.* Consecutive snapshots overlap by 50 minutes; all intervals use day-block resampling for this reason, but the effective sample size is closer to the number of days than to the number of snapshots.
- *Indices versus path.* Juliusruh is about 900 km south-west of the transmitter and not on the path to North America; K and A are planetary values, and SFI is daily. The indices are proxies.
- *SNR.* Reported SNR is estimated by the receiving station and depends on its noise floor and software; only medians over many stations are interpreted.

6 Data quality and software notes

Two issues were found while preparing the study and are relevant to anyone reading the archive.

Under-reported Best DX. The logger's per-transmission record (used for the "Best DX" figure on the live page) was written from a single poll six minutes after the transmission. Distant stations upload to WSPR Net later, so the farthest reporter of a transmission was often missing; comparing the two databases day by day showed that on 67 of 163 days (41 %) the farthest reporter in the replay data was farther than the stored value (largest difference on 2026-10-07: 4 553 km stored against 15 614 km). The poll now covers the last 45 minutes and raises the stored value as later reports arrive; days already stored were not corrected, and this study uses the replay database.

Spurious reporter. One callsign–grid pair carried a distance that contradicted its own grid (Section 2.5); this is the kind of entry that the cleaning step is meant to catch, and a similar check could be added to the live page.

7 Conclusions

- The band is open (median at least 50 stations) from 7 to 20 UTC, with a median of about 78 stations at midday; the long-distance windows lie at the edges of the day, not in its middle.
- North America is reached most often in the morning (10–13 UTC), Africa and Oceania in the afternoon and evening (16–19 UTC); the farthest reception was 15 619 km.
- 79 % of days had a reporter beyond 6 000 km with 200 mW and a car-roof whip.
- Within the day, the MUF matters more than anything else the logger records; K and A matter most for the longest paths; the daily solar flux has no detectable effect.
- Autumn was far better than summer in this record, independently of the solar flux.

A Reproducing the analysis

The figures, tables and every number in the text are produced from the two SQLite databases by the script `doc/src/analyze.py` (Python 3 with NumPy and Matplotlib; the standard library for everything else). It writes `numbers.tex`, one `tab_*.tex` file per table and the PDF figures to `doc/src/out`; the paper is then built with `pdflatex propagation_study.tex` (twice). The databases themselves are not part of the repository.

Version history

Version	Date	Changes
1.0	6 November 2026	First version: 2026-05-08 to 2026-11-06, 18 867 snapshots.

References

[1] WSPRnet — the Weak Signal Propagation Reporter Network. `https://wsprnet.org`

[2] wspr.live — ClickHouse interface to the WSPRnet spot database. `https://wspr.live`

[3] J. Taylor, K1JT, *WSJT-X User Guide* (WSPR mode) and *WSPR 2.0 User's Guide*. `https://wsjt.sourceforge.io`

[4] ZachTek, *WSPR Desktop Transmitter* documentation. `https://www.zachtek.com/1012`

[5] Global Ionospheric Radio Observatory, Digital Ionogram Database (DIDBase). `https://giro.uml.edu/didbase/`

[6] Leibniz Institute of Atmospheric Physics, Juliusruh Space Weather Monitor. `https://ionosonde.iap-kborn.de/actuellz.htm`

[7] NOAA Space Weather Prediction Center, solar and geophysical indices. `https://www.swpc.noaa.gov`

[8] N0NBH, solar-terrestrial data feed. `https://www.hamqsl.com/solar.html`